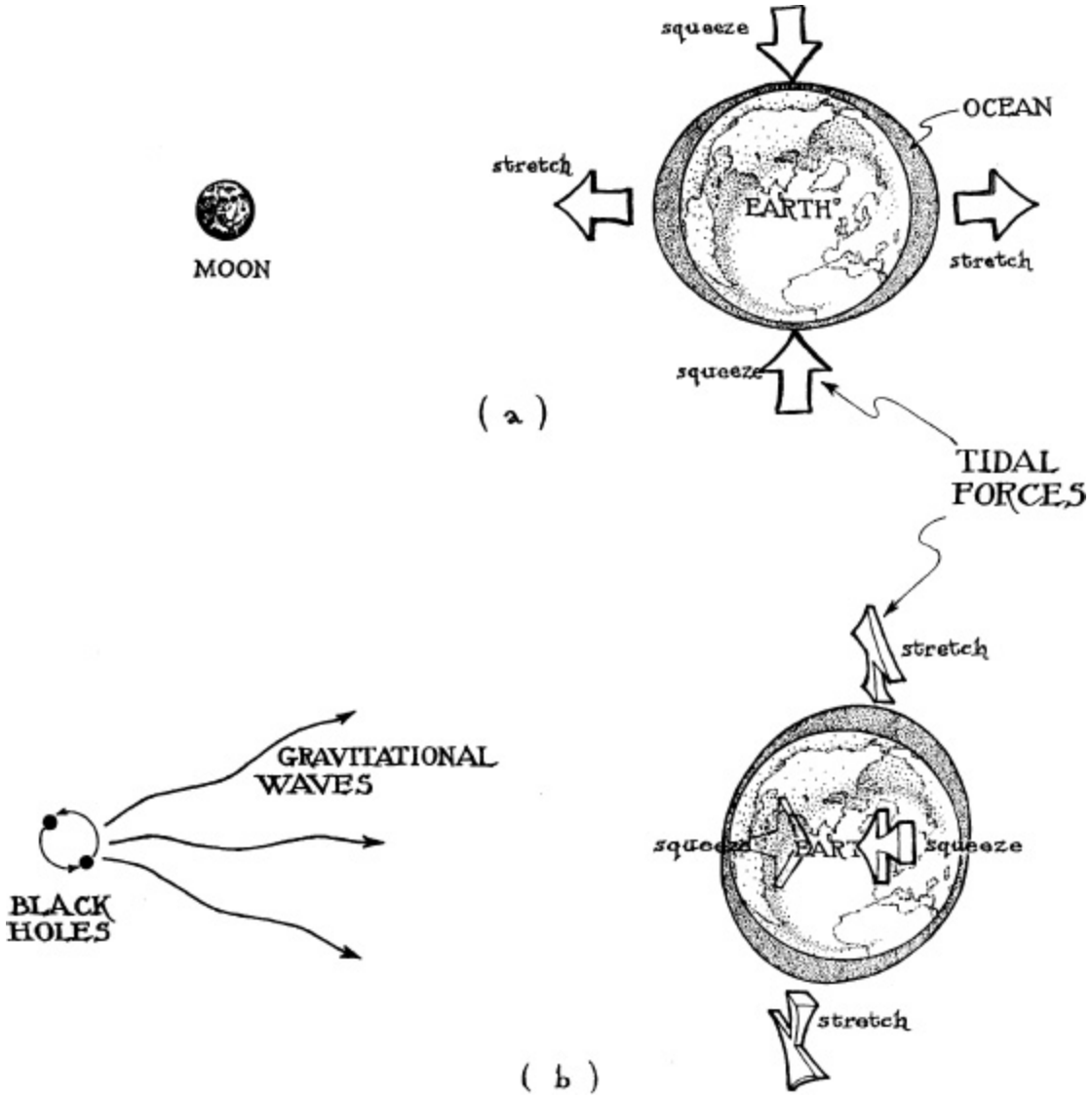


10.3 The tidal forces produced by the Moon and by a gravitational wave. (a) The Moon's tidal forces stretch and squeeze the Earth's oceans; the stretch is longitudinal, the squeeze is transverse. (b) A gravitational wave's tidal forces stretch and squeeze the Earth's oceans; the forces are entirely transverse, with a stretch along one transverse direction and a squeeze along the other.

The third difference between the Moon's tides and those of a gravitational wave is their size. The Moon produces tides roughly 1 meter in size, so the difference between high tide and low tide is about 2 meters. By contrast, the gravitational waves from coalescing black holes should produce tides in the Earth's oceans no larger than about 10^{-14} meter, which is 10^{-21} of the size of the Earth (and 1/10,000 the size of a single atom, and just 10 times larger than an atom's nucleus). Since tidal forces are